HYUNDAI-ROTEM (H-R) SUBMITTAL REVIEW FORM

H-R Submittal Reference:	HRU-TRN-000600 Design Data Package: #10 (FDR)	**Date Submitted:** 18th July 2012	**DTS Review #** DTS-HRU-R0426

Doc No.	Title	Rev	CDRL Ref.	Spec Ref
See Below	Communications & CCTV FDR Design Package (Detailed Below)	3	9-16.R13F	RTD Concession Agreement Attach 7 Part C Section 12 Rolling Stock Contract Part I Section 3.17

DTS/DTP Submittal Review Comments :

The following revised and/or additional documents were submitted for review as part of the Communications & CCTV FDR Design Package in response to DTS review DTS-HRU-R0382.

- HR Document REDE114273, Rev 3 – System Functional Description for Communications System
- HR Document REDE114275, Rev 3 – System Functional Description for CCTV System
- HR Document REDE114276, Rev 3 – Operational Scenario for CCTV System
- HR Document REDE113108, Rev 6 – Technical Description for CCTV System
- HR Document REDE113112, Rev 5 – Technical Description for Communications System
- Inter-M Document IM-FE-TD-010, Rev. 3 – Monitor GUI of CCTV
- Several Communications & CCTV System Drawings

As stated in DTS-INFML-000942, the CCTV FDR Design Package cannot be accepted until the OCC Viewing Software / Compatibility with the Wayside CCTV System issue is resolved to DTS satisfaction.

☑ *See continuation sheet*

Approval Status *	A	B	C	IO
PDR				
FDR		X		

Reviewer's Signature:		**Review Date:**	3rd August 2012

**** Key to Approval Status Codes:***

A = Proceed
B = Revise and Resubmit. Work may proceed subject to incorporation of changes indicated.
C = Do not proceed. Change as noted and resubmit.
IO = Data accepted for information only.

DTS/DTP Submittal Review Comments Continued:

All DTS comments submitted with DTS-HRU-R0382 have been addressed by HR and can be considered closed except for the following:

Comment 2.35 – It appears that changing the time to 90-seconds is making the warm start requirement even more restrictive than the original 3-minutes. Why does the warm start need to be linked to a time? A warm start should occur whenever power has not been removed from the communications system. Perhaps a block/flow diagram would help with our understanding.

Comment 3.3 – Please find attached a marked-up CCP User Interfaces document reflecting the recent change from TRAIN numbers to ROUTE numbers. Please update the CCP User Interfaces document and other documents that show screen shots of the CCP.

Comment 2.37, Bullet #3 – An updated Interior Speaker drawing was not included in the revised package.

Comment 2.37, Bullet #7 – Woojin Document 423-309, Section 3.5.2.3 requires updating to the latest PEI call announcement.

The following action items remain open from the FDR meeting and Communications/CCTV Design Interface Workshops with Wayside meetings:

- RS-MM-054-17, Action Item #719 (DTP action; this should be discussed at a future workshop or bench test with XO Rail)
- RS-MM-084-5, Action Item #996 (DTP action; this should be discussed at a future workshop or bench test with XO Rail)
- RS-MM-084-12, Action Item #1004 (DTP action; this should be discussed at a future workshop or bench test with XO Rail)
- RS-MM-085-63, Action Item #1035 (DTP action; this should be discussed at a future workshop or bench test with XO Rail)

Open action items related to the 2nd Communications & CCTV Systems FAI/Type Test:

- Action Item #1397 (HR action)
- Action Item #1400 (DTO action)
- Action Item #1401 (DTO action)

DTS does not have any new comments.

THE DRAWING(S) AND THE INFORMATION CONTAINED HEREIN ARE THE PROPERTY OF **WOOJIN INDUSTRIAL SYSTEMS CO., LTD.** THEY SHALL NOT BE DISCLOSED, REPRODUCED OR USED, IN ANY MANNER WITHOUT THE PRIOR WRITTEN CONSENT OF **WOOJIN INDUSTRIAL SYSTEMS CO., LTD.**

DESCRIPTION
OF
CCP USER INTERFACES

DISTRIBUTION	

WOOJIN	Description of CCP User Interfaces	Document no.	42A-2012-0055
		Issued date	June 20, 2012
		Rev. no.	3

REVISION RECORD

Rev. No.	PAGE	Details		Date
		FROM	TO	
0		First Issued		March 21, 2012
1	ALL	Previous Figures	Revised Figures	May 4, 2012
2	ALL	Previous Figures	Revised Figures	June 4, 2012
3	ALL	Previous Figures	Revised Figures	June 20, 2012

Table of Contents

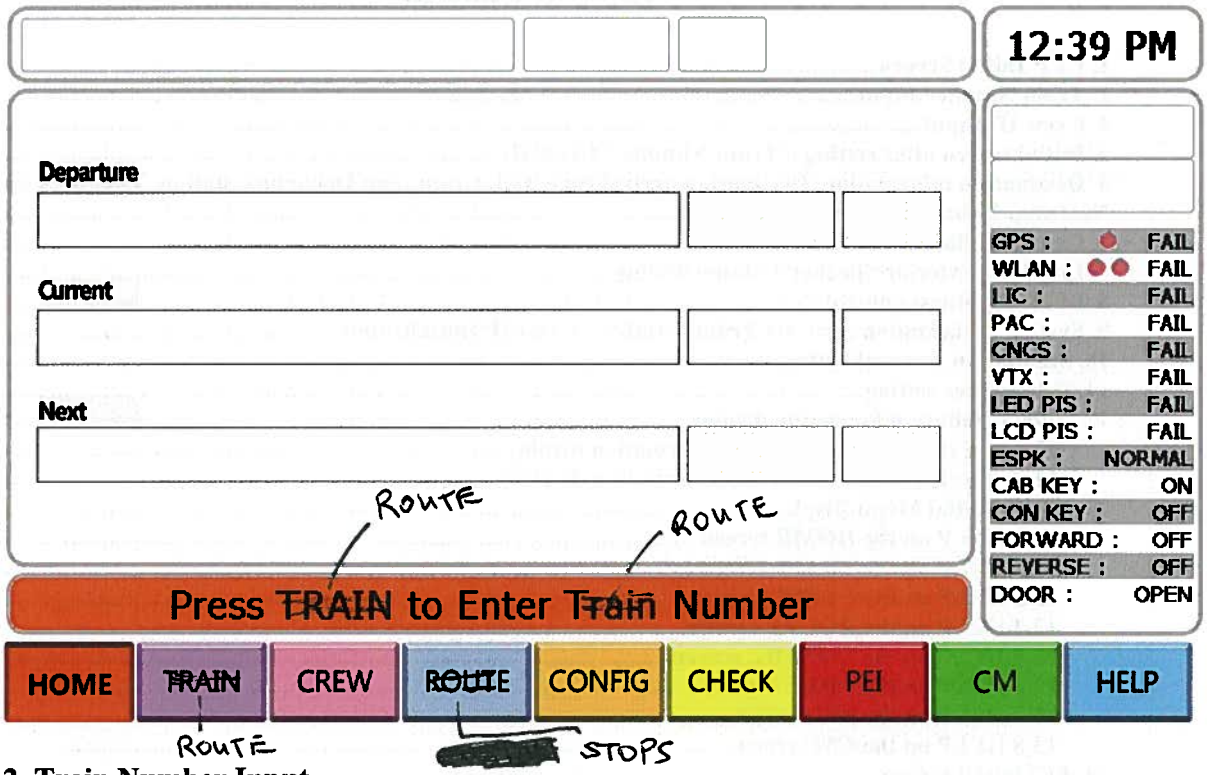
1. CCP Initial Screen 3
2. Train Number Input 3
3. Crew ID Input 4
4. Initial screen after setting a Train Number / Crew ID 5
5. Destination information, Destination arrival time, Delay time, and Departure station, This stop, and Next stop Information Display 6
6. Configuration Initial Screen 7
7. Interior / Exterior Speaker Volume Setting 8
8. LCD brightness control 8
9. System initialization : pre-set Train Number / Crew ID initialization 9
10. Simulation Interval Setting 10
11. C/I number setting 11
12. Vehicle failure information display 12
- **12.1 Married A vehicle failure information display** 13
- **12.2 Married B vehicle failure information display** 14

13. HELP Initial Menu Display 15
- **13.1 HELP on the HOME screen** 15
- **13.2 HELP on the TRAIN screen** 16
- **13.3 HELP on the CREW screen** 17
- **13.4 HELP on the ROUTE screen** 17
- **13.5 HELP on the CONFIG. screen** 18
- **13.6 HELP on the CHECK screen** 19
- **13.7 HELP on the PEI screen** 19
- **13.8 HELP on the CM screen** 20

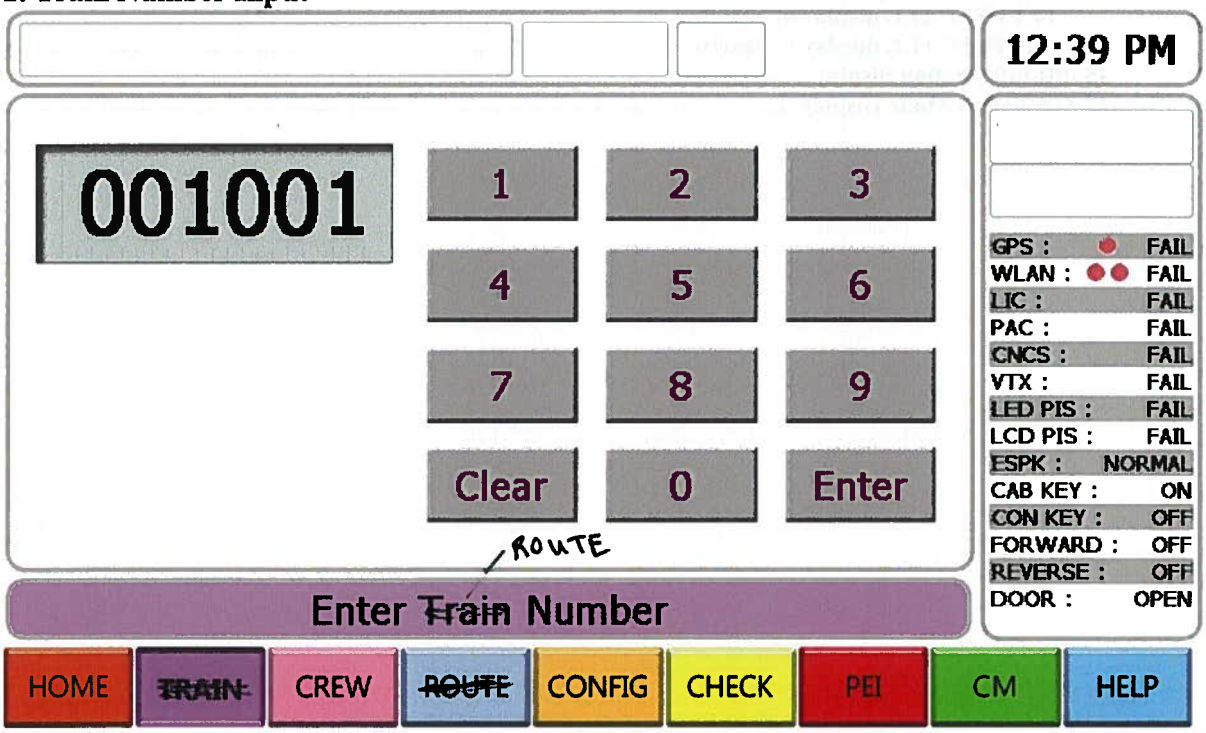
14. PEI initial screen 20
- **14.1 PEI CALL display of Married A** 22
- **14.2 PEI CALL display of Married B** 23

15. PR information display 23
16 Simulation Mode Display 24

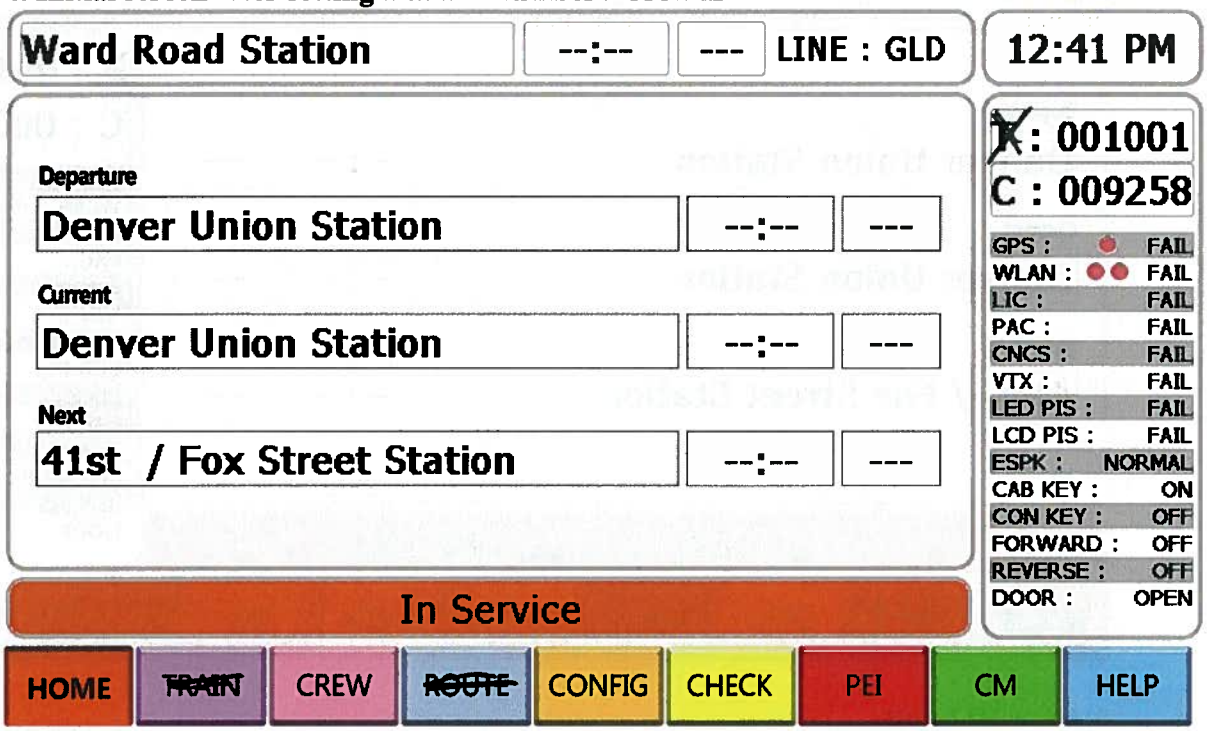
1. CCP Initial Screen

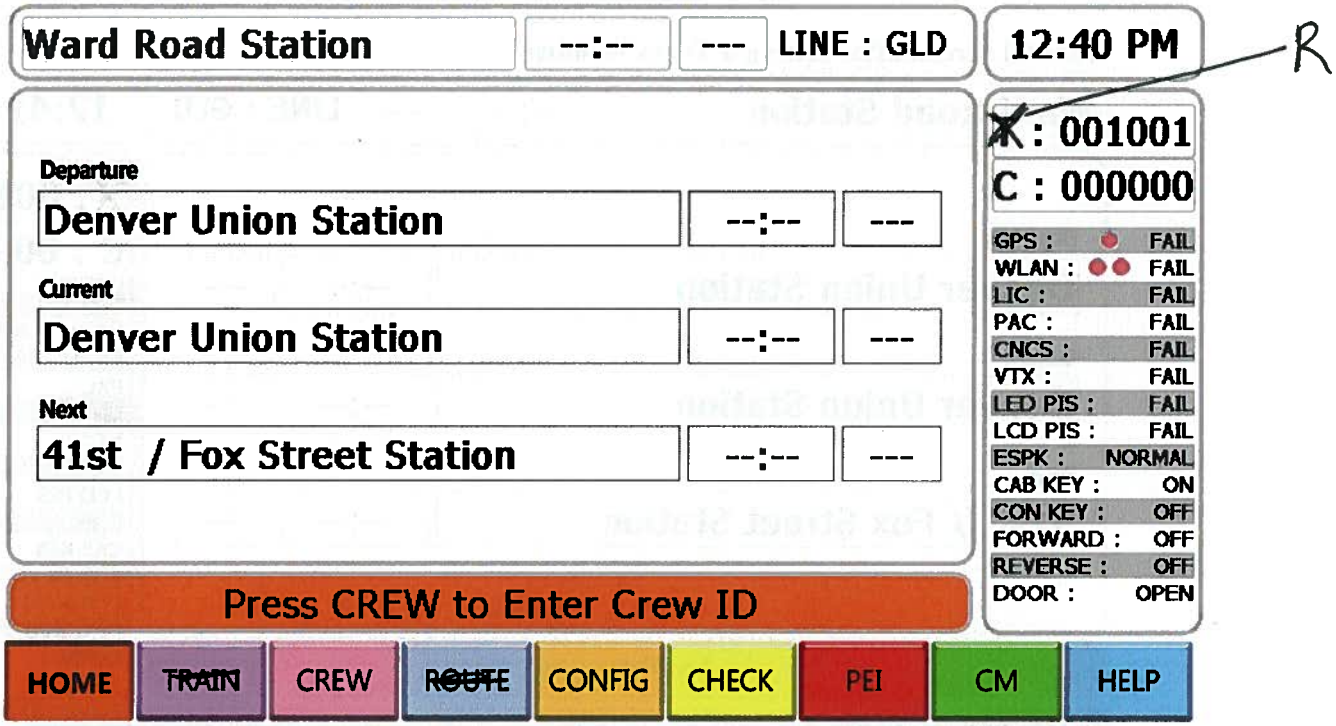


2. Train Number Input

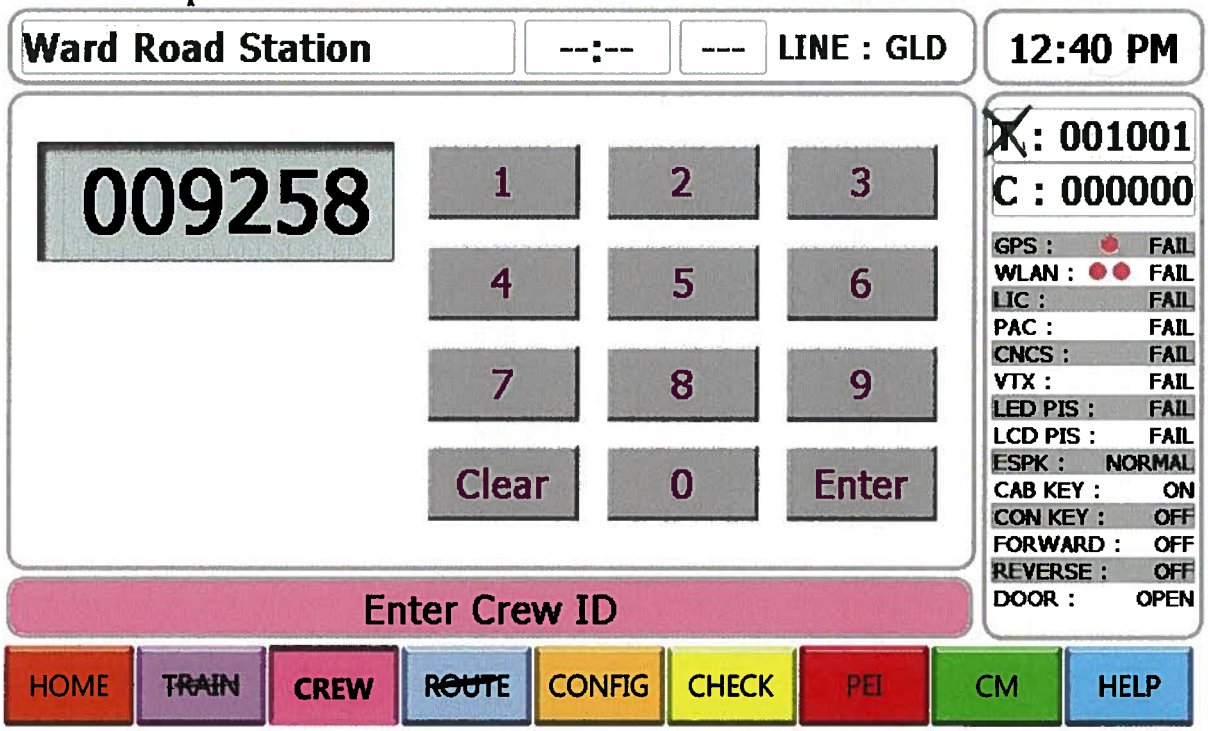


4. Initial screen after setting a Train Number / Crew ID

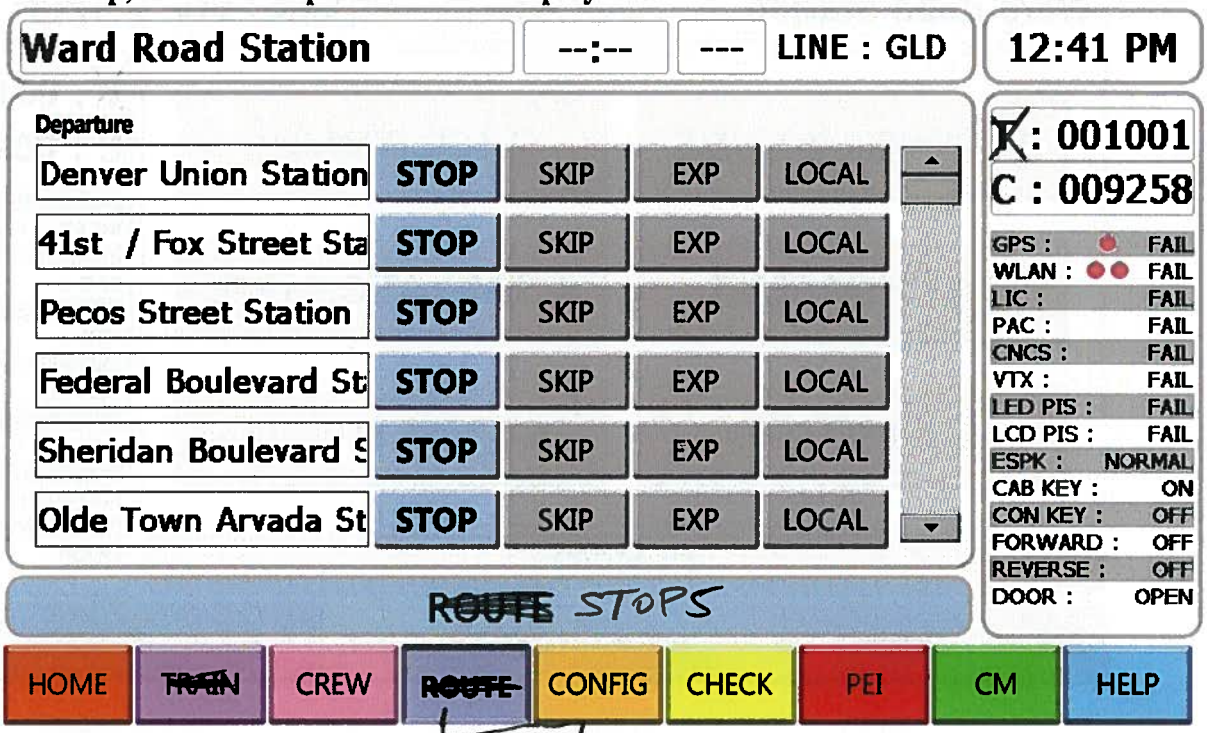




3. Crew ID Input



5. Destination information, Destination arrival time, Delay time, and Departure station, This stop, and Next stop Information Display

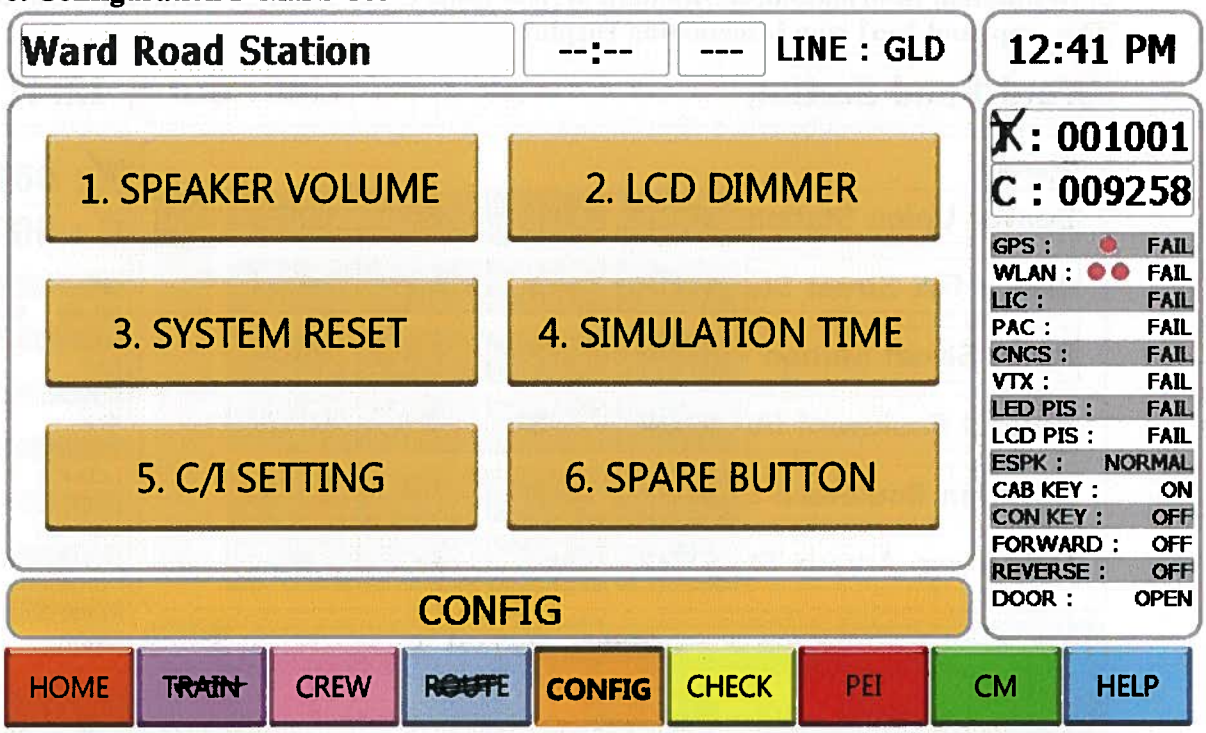


- Route setting STOPS

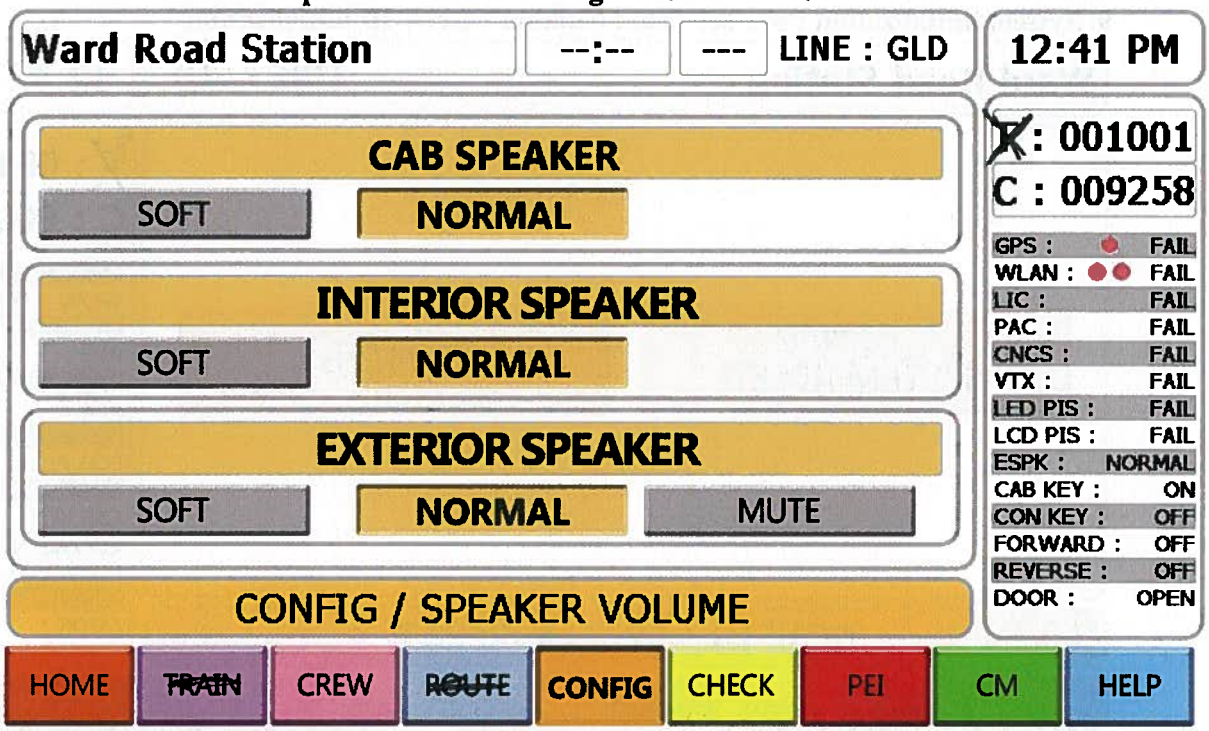
STOPS

STOP :It transmits an automatic announcement.
SKIP : It does not transmit the automatic announcement.
EXP : It is considered as "SKIP" from the terminal to the selected station.
LOCAL : It is considered as "STOP" from the terminal to the selected station.

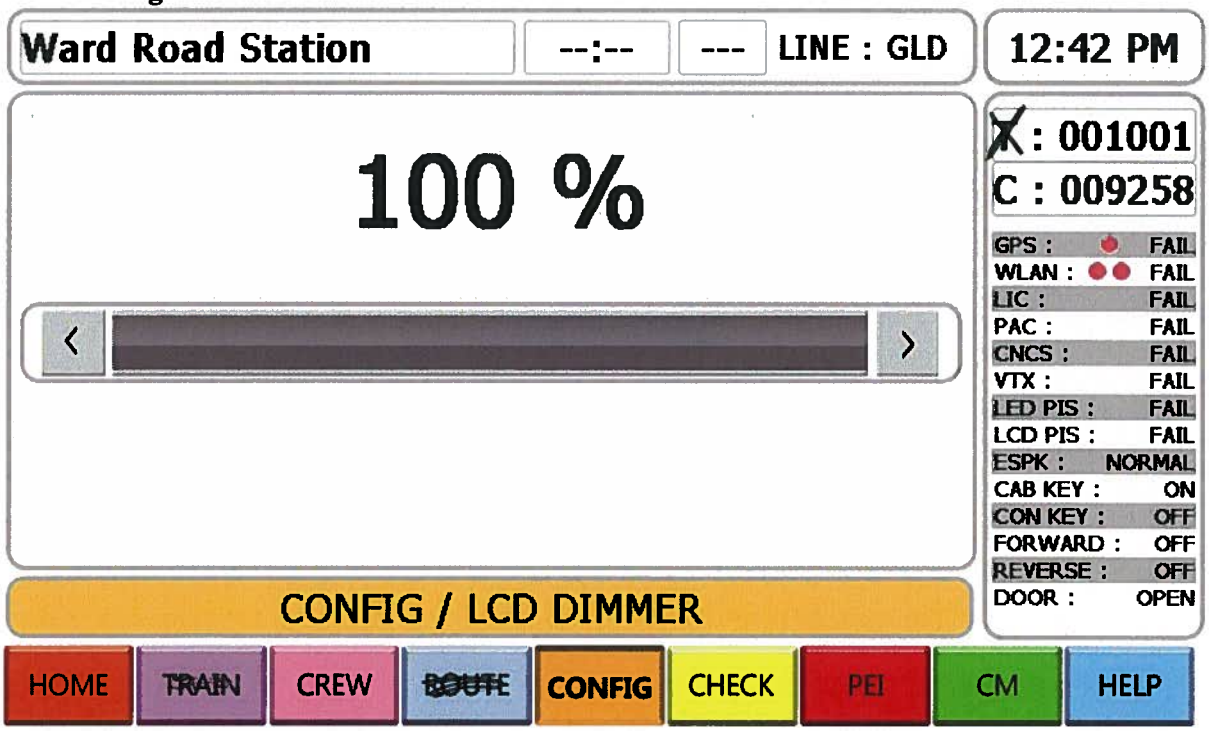
6. Configuration Initial Screen



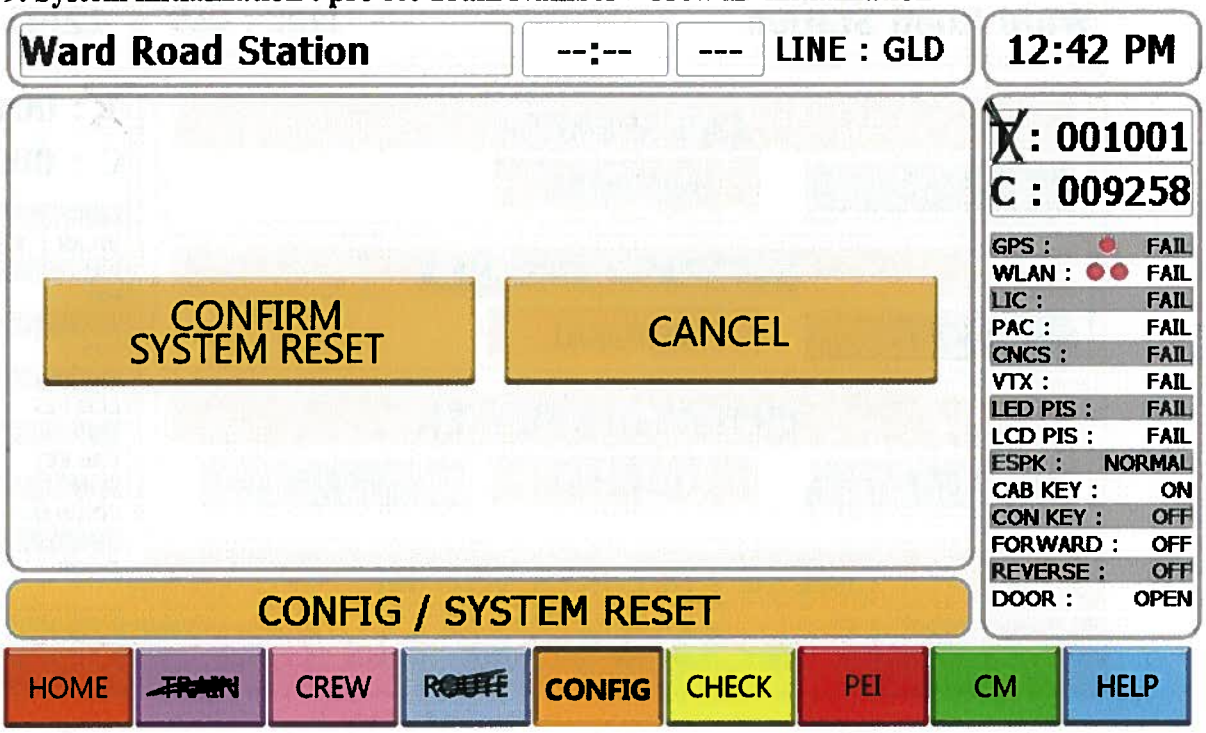
7. Interior / Exterior Speaker Volume Setting * to be password protected



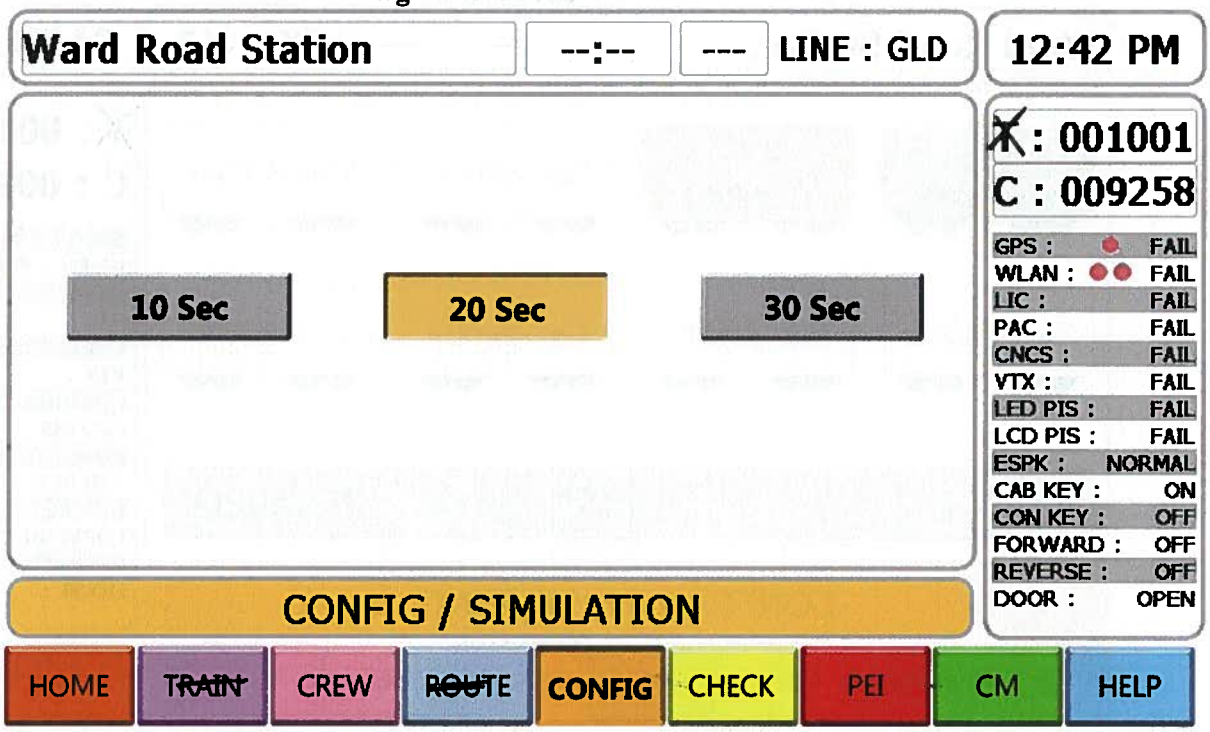
8. LCD brightness control



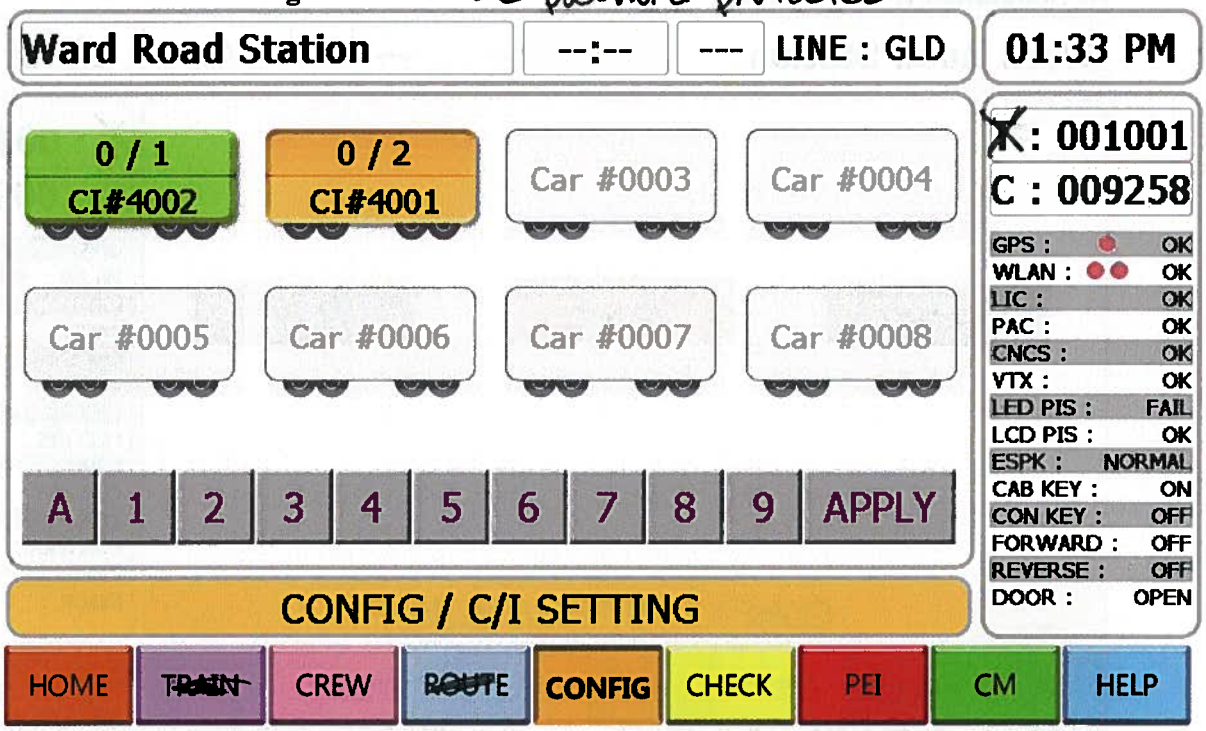
9. System initialization : pre-set Train Number / Crew ID initialization



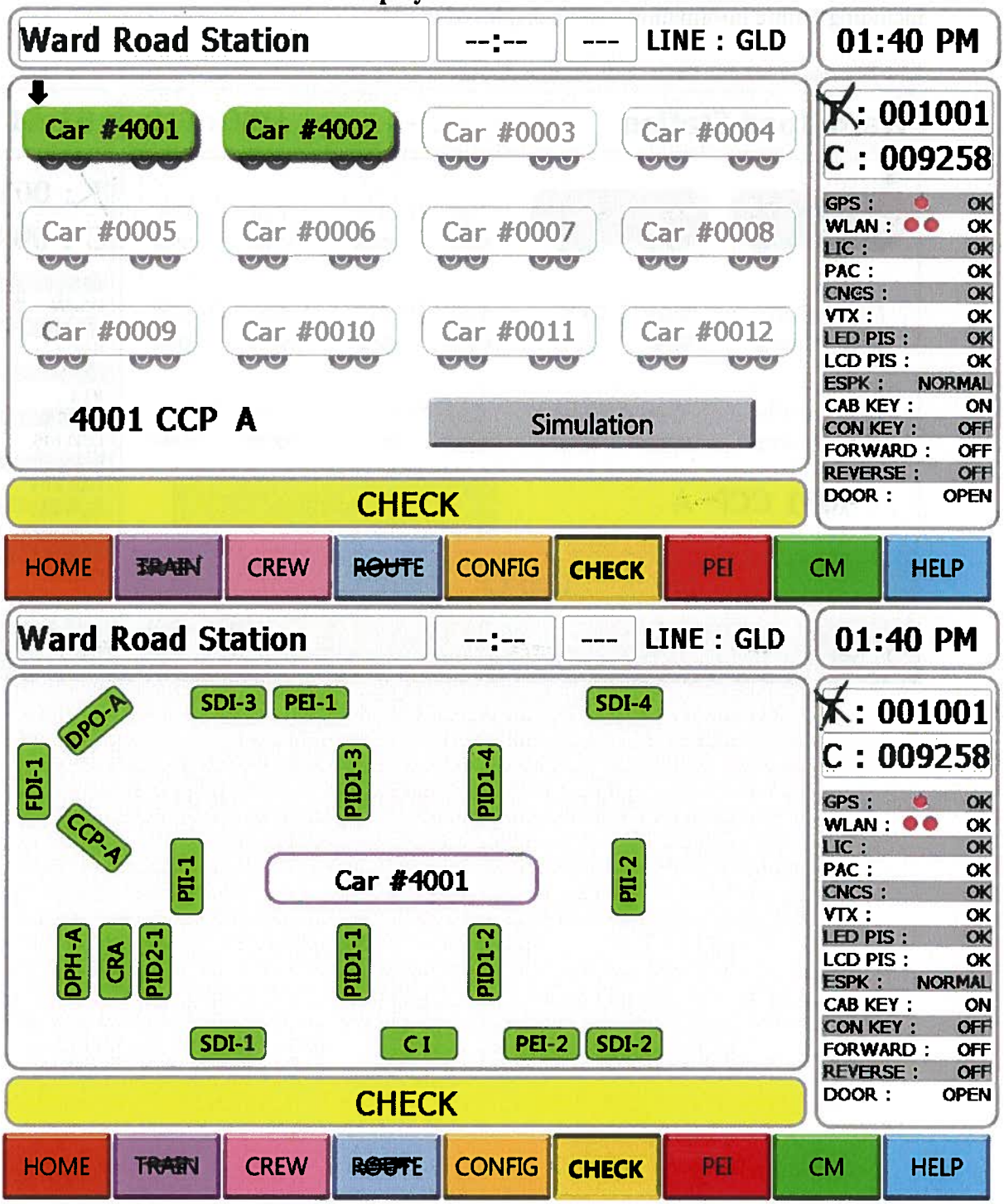
10. Simulation Interval Setting



11. C/I number setting * to be password protected.

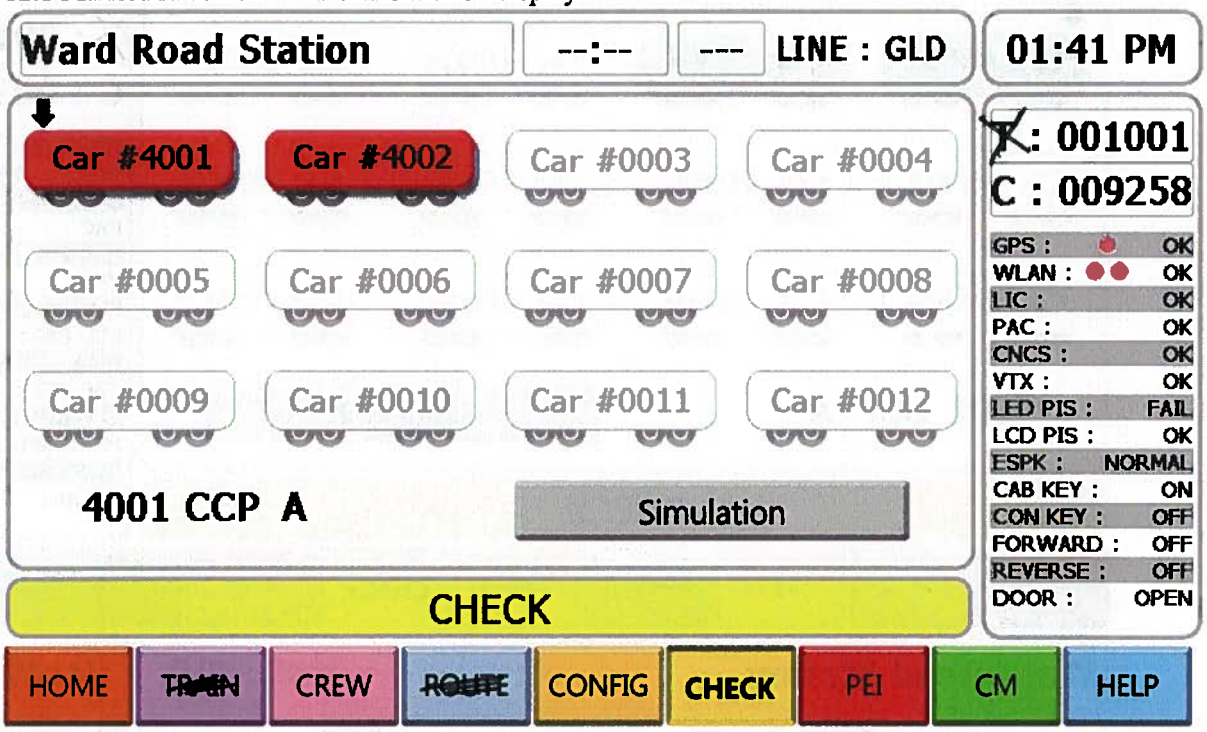


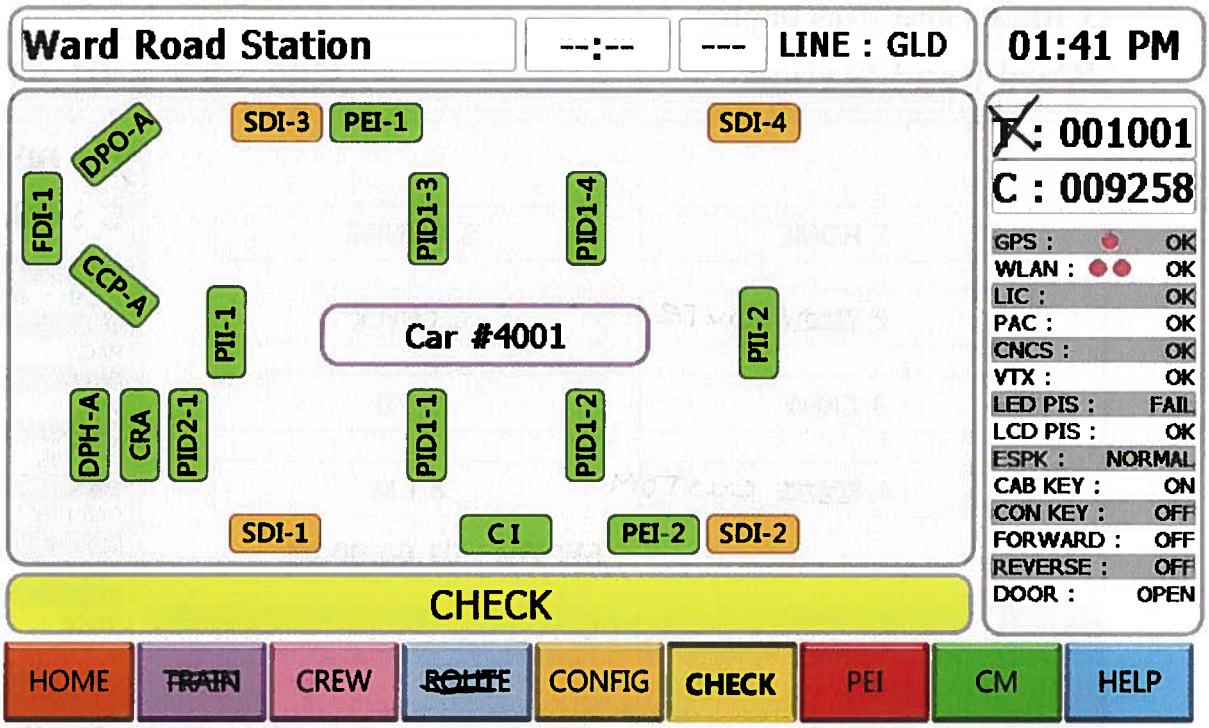
12. Vehicle failure information display



A vehicle without any failure information will be displayed with green color, but a vehicle including failure information will be displayed in red.

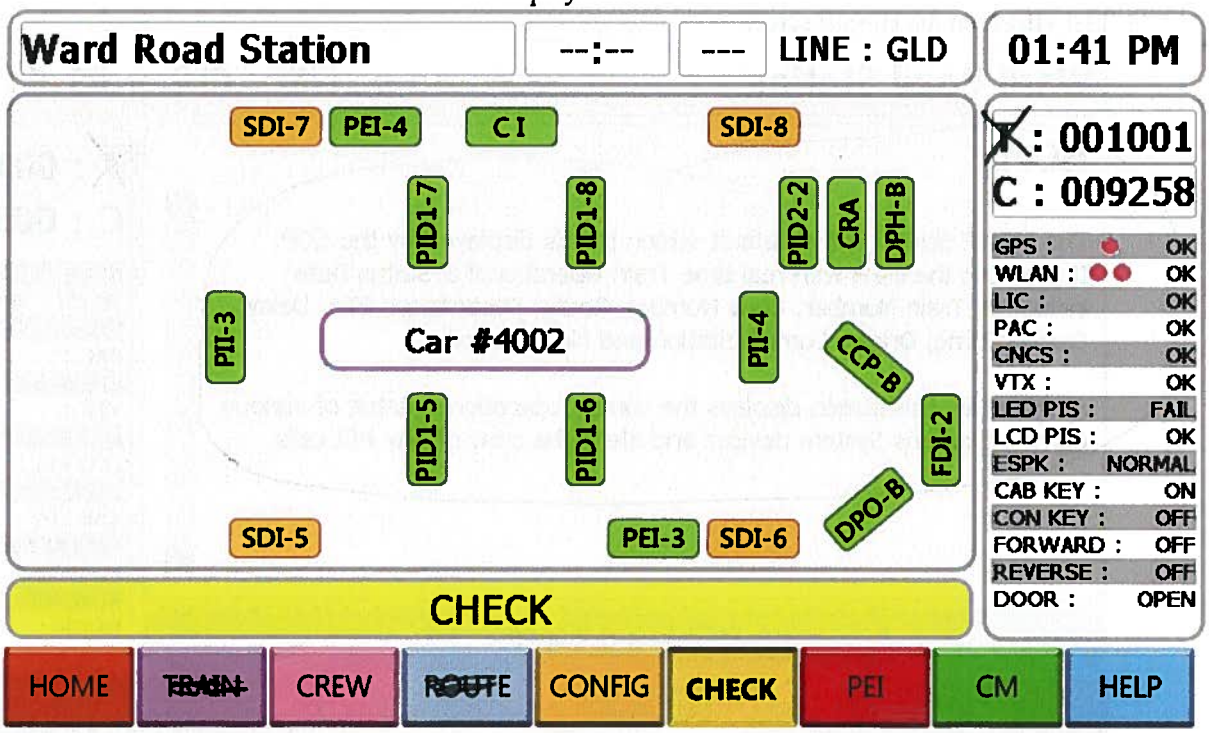
12.1 Married A vehicle failure information display



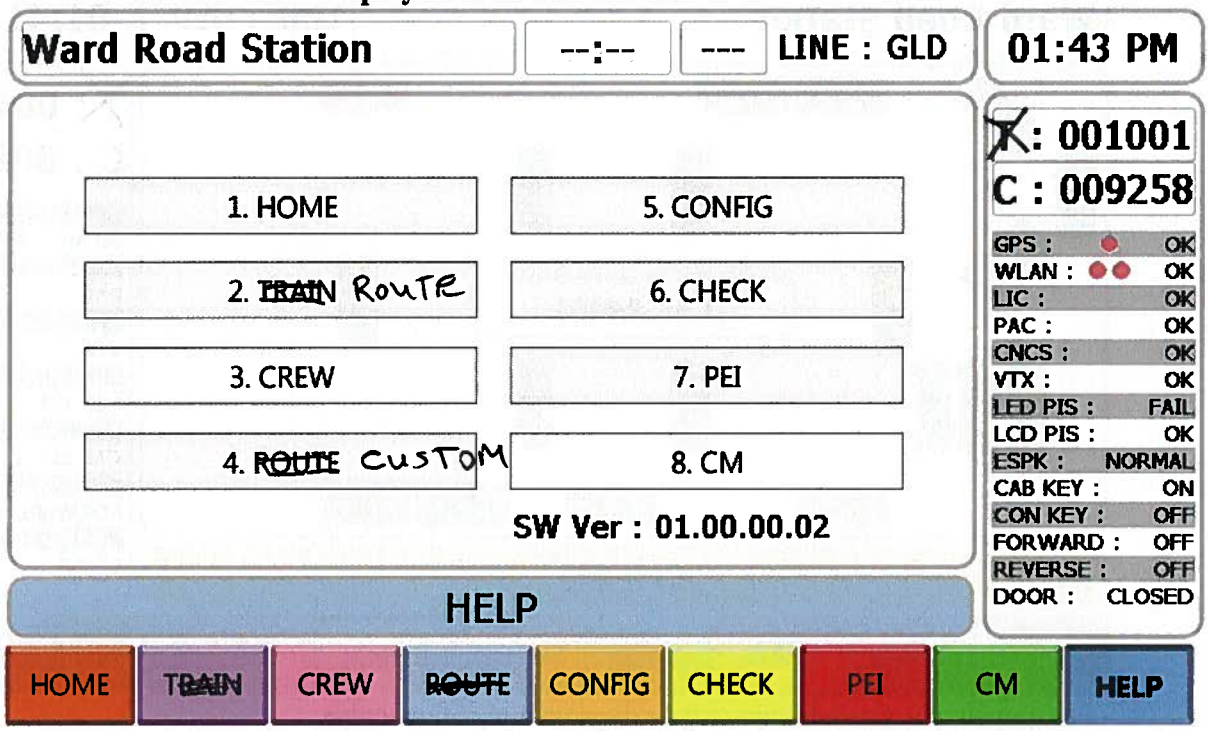


- Equipment which has failure will be displayed in yellow color.

12.2 Married B vehicle failure information display

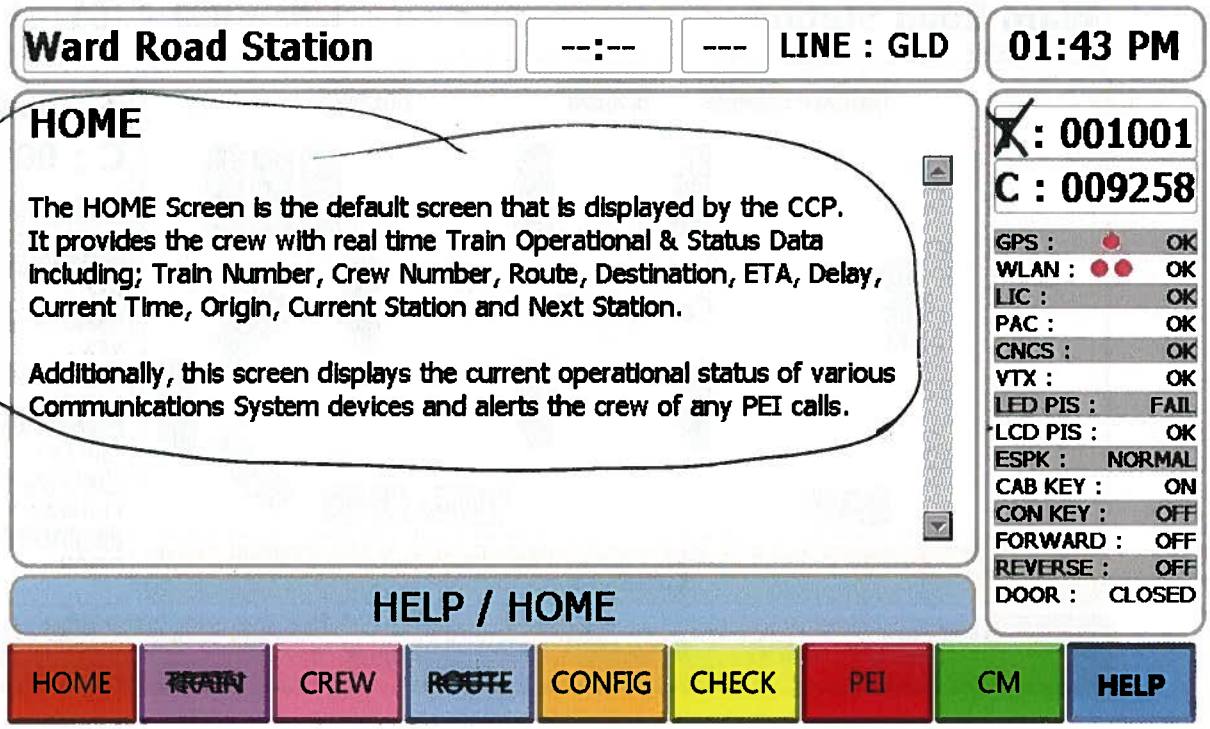


13. HELP Initial Menu Display

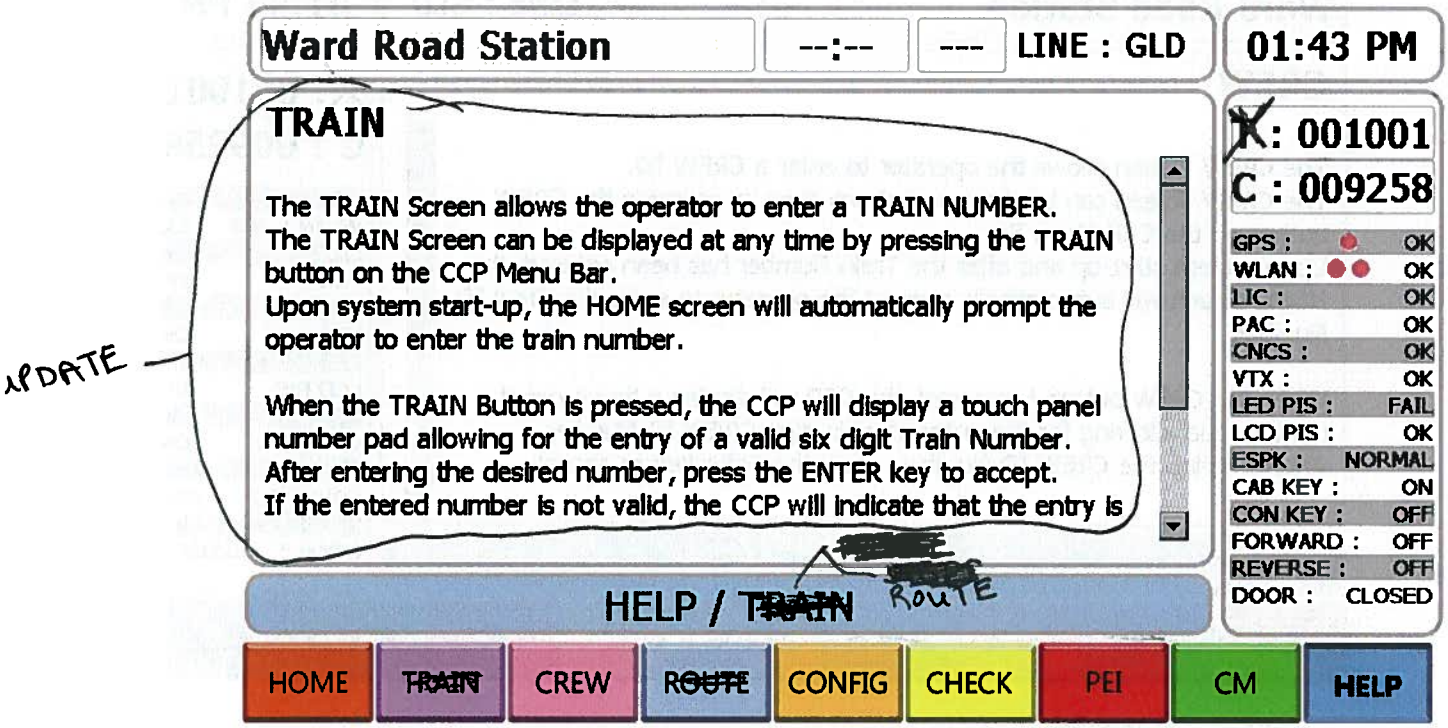


13.1 HELP on the HOME screen

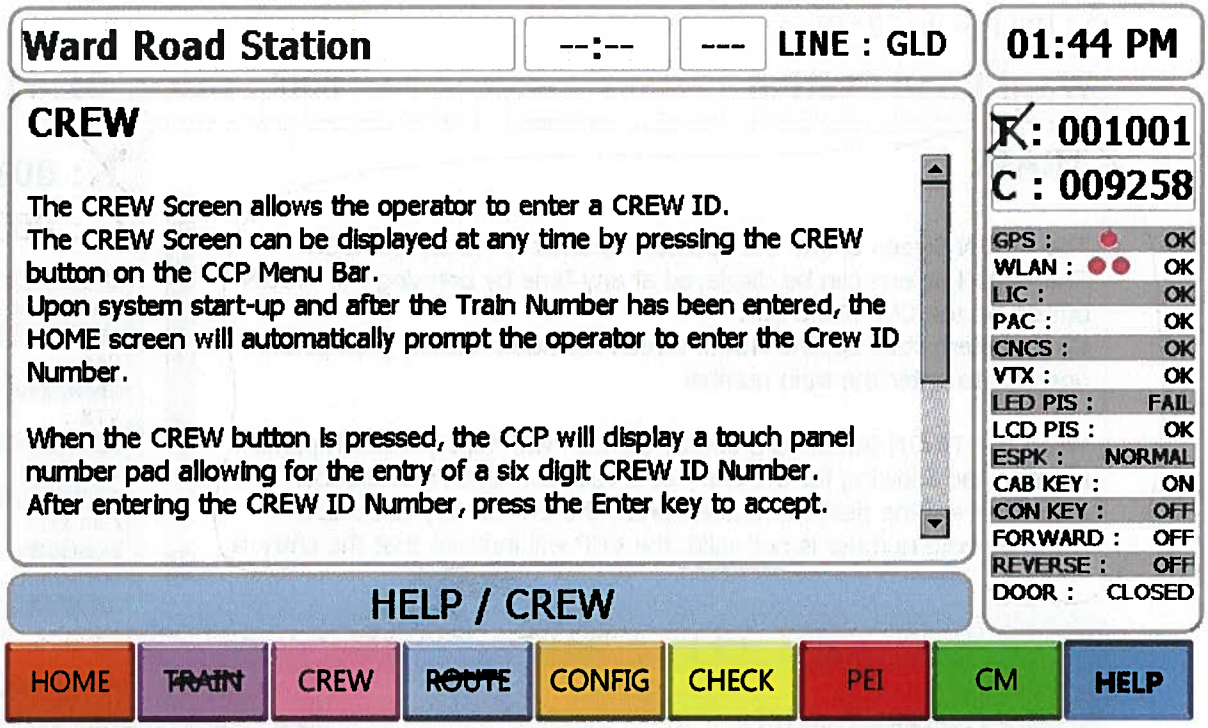
UPDATE



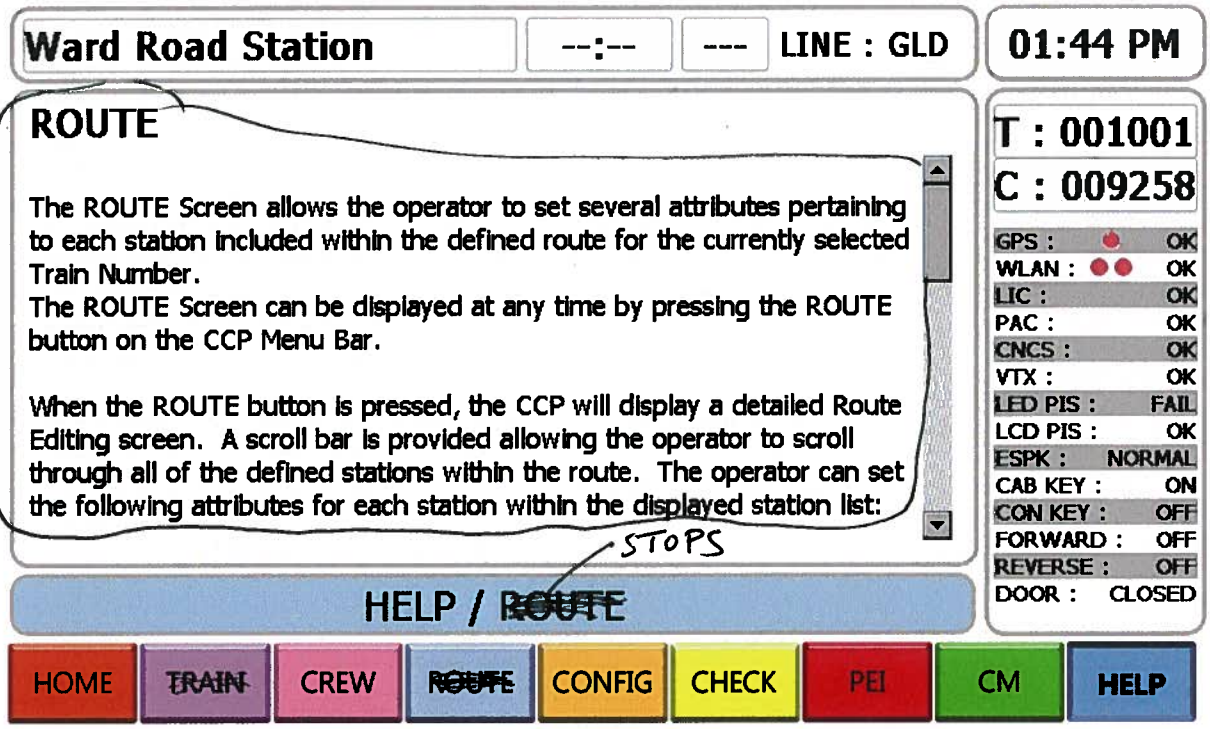
13.2 HELP on the TRAIN screen



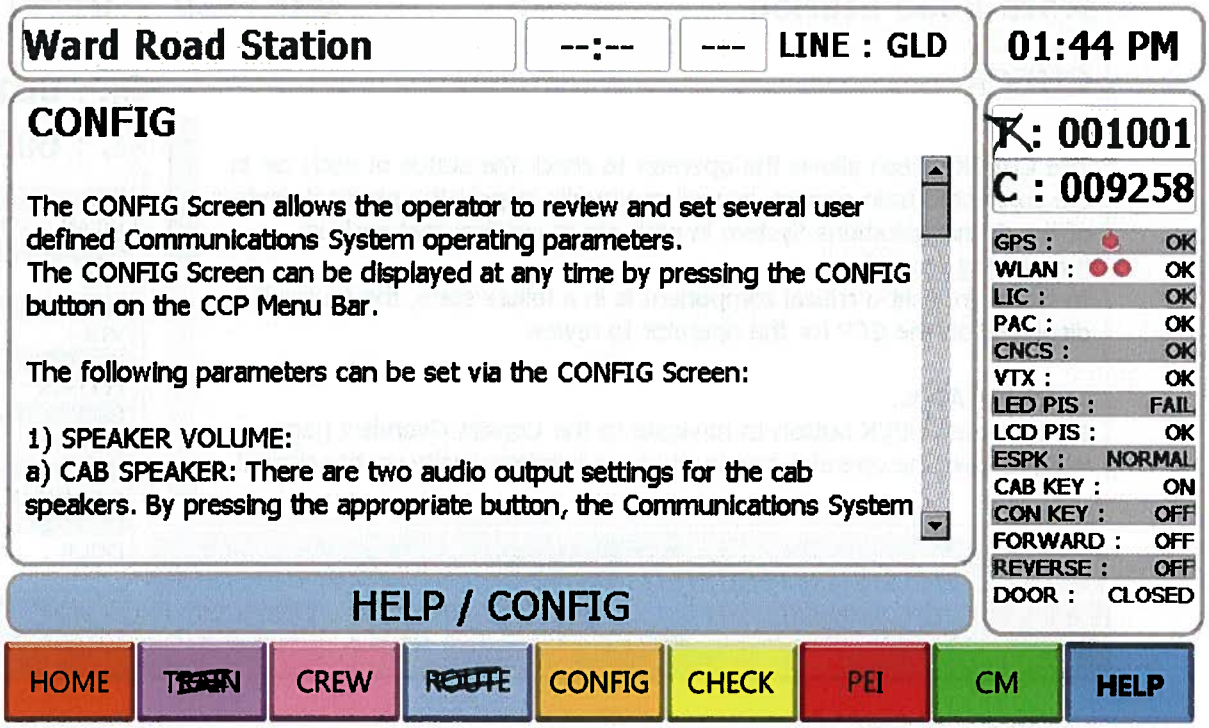
13.3 HELP on the CREW screen



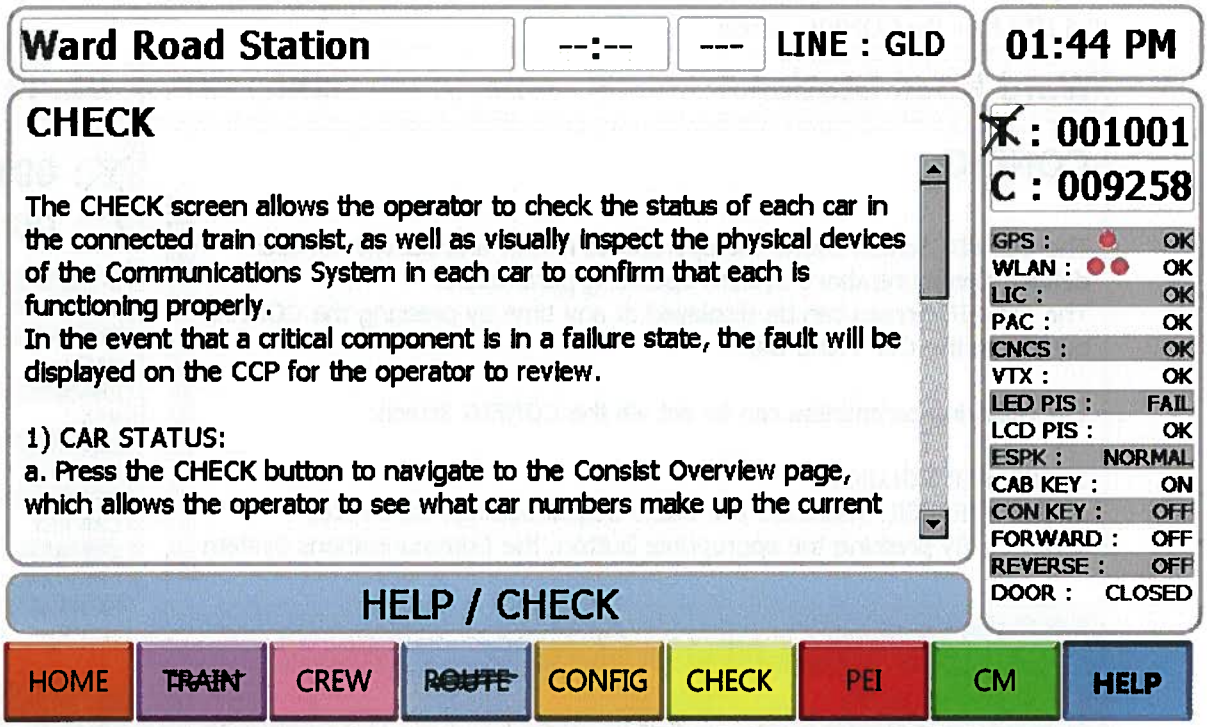
13.4 HELP on the ROUTE screen



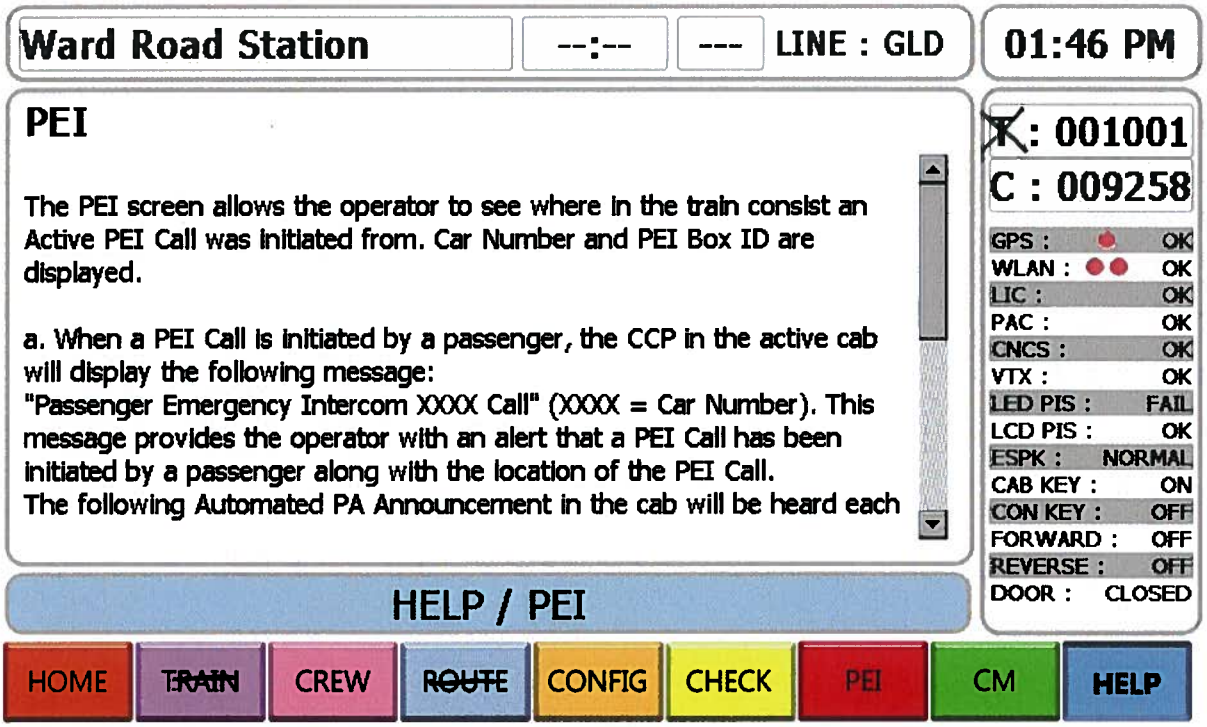
13.5 HELP on the CONFIG. screen



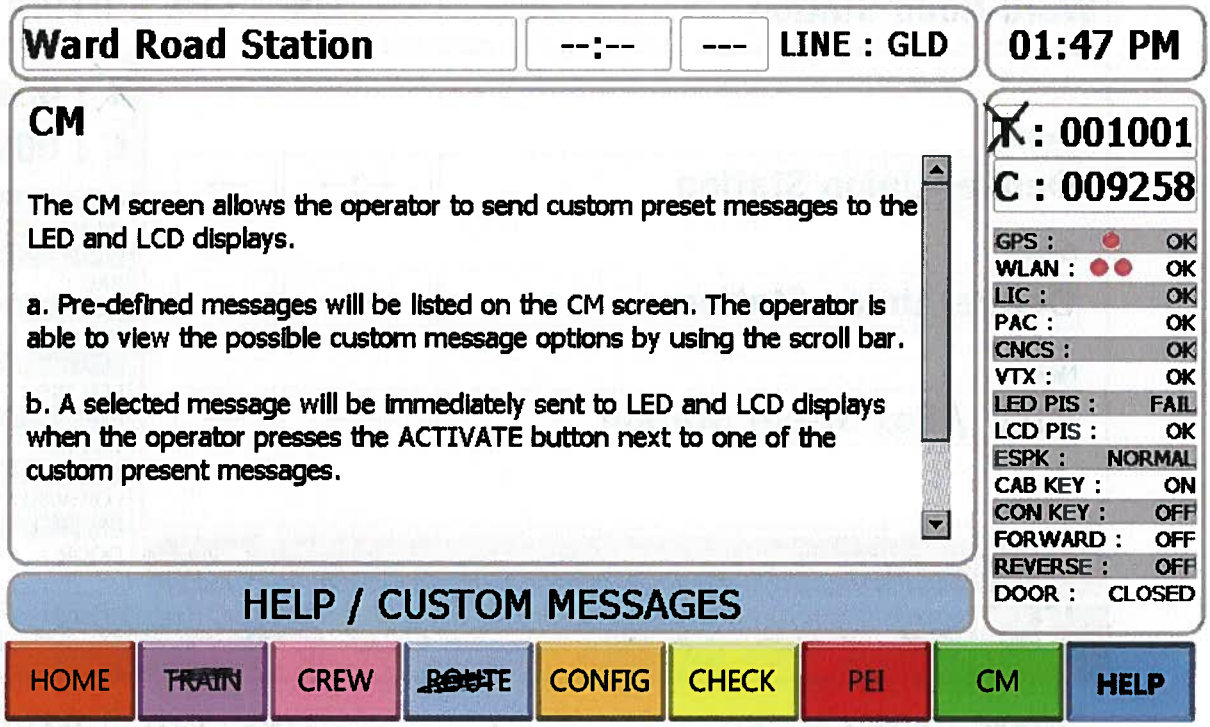
13.6 HELP on the CHECK screen



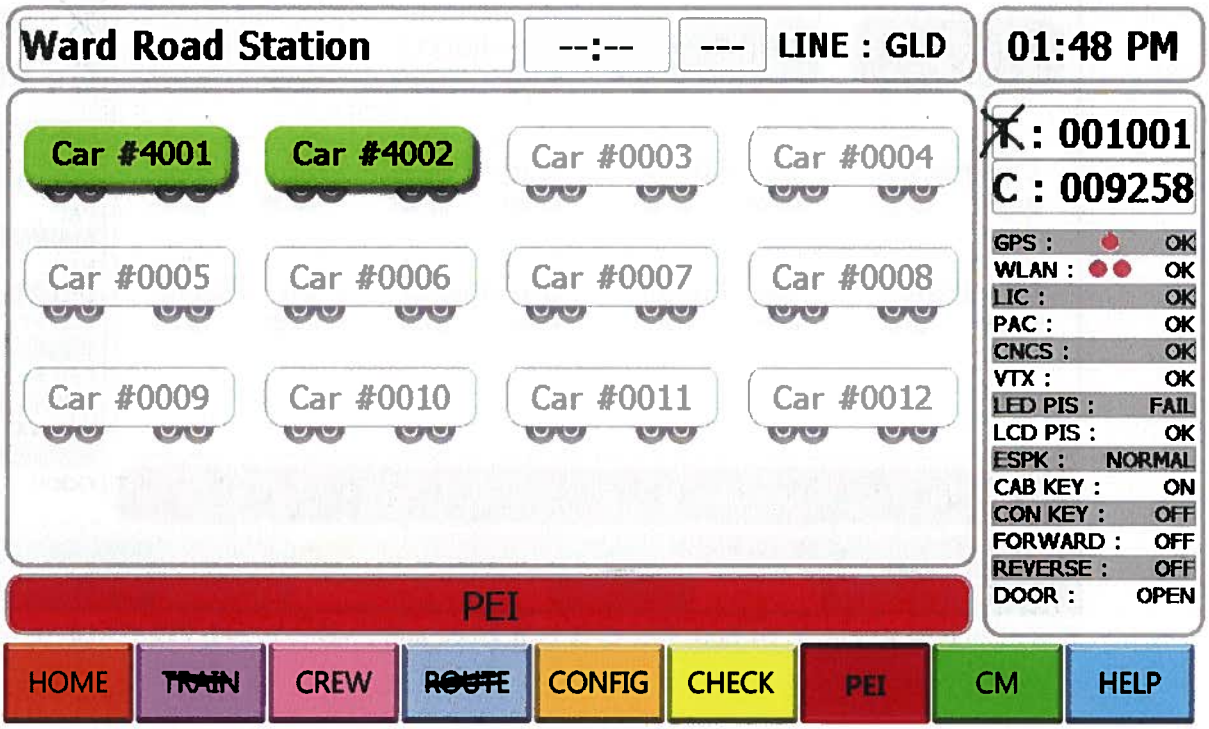
13.7 HELP on the PEI screen

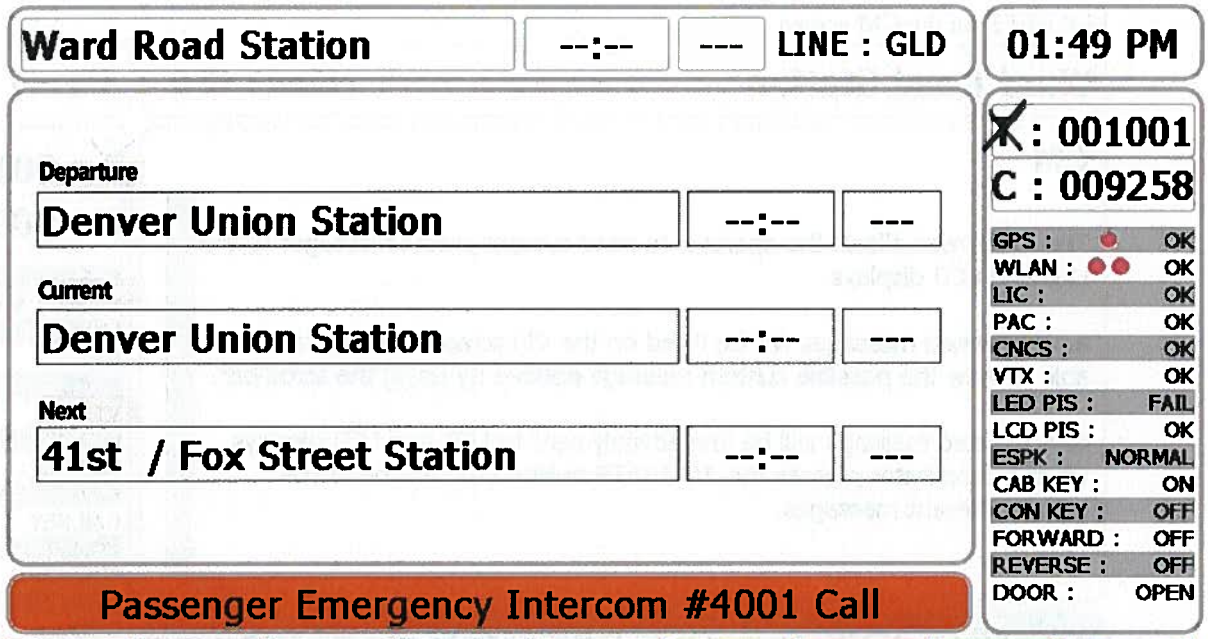


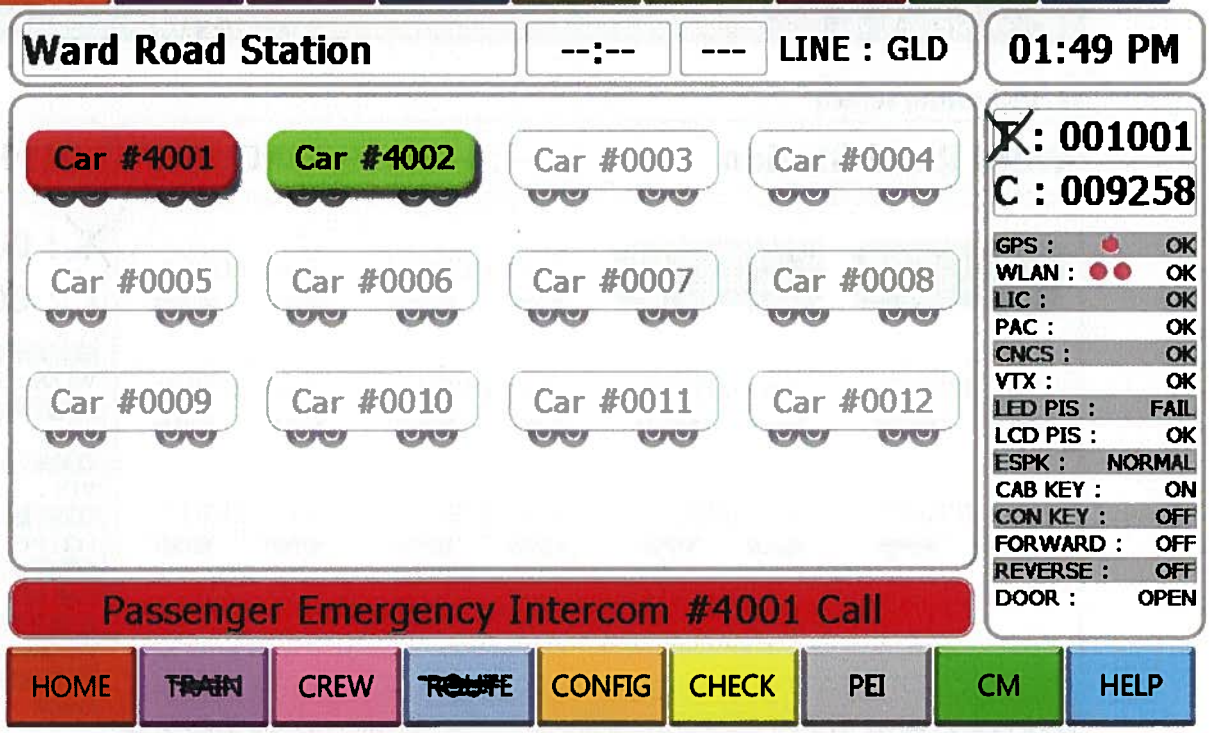
13.8 HELP on the CM screen



14. PEI initial screen

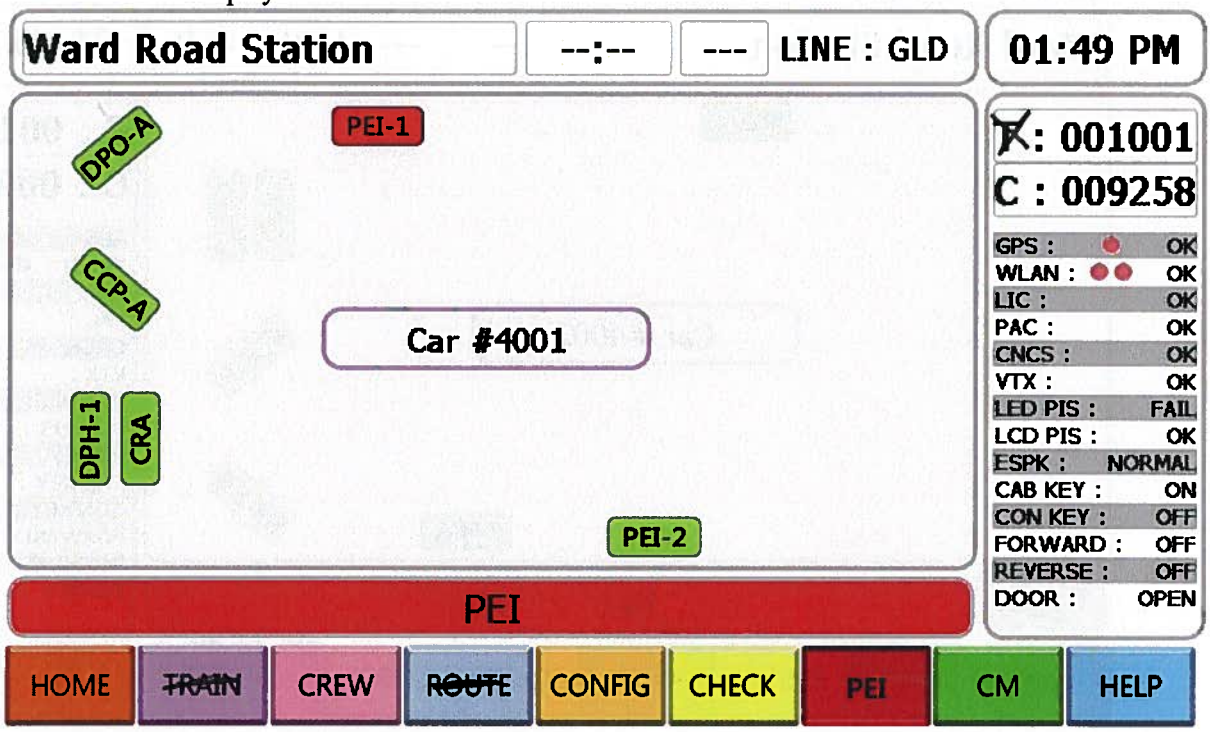






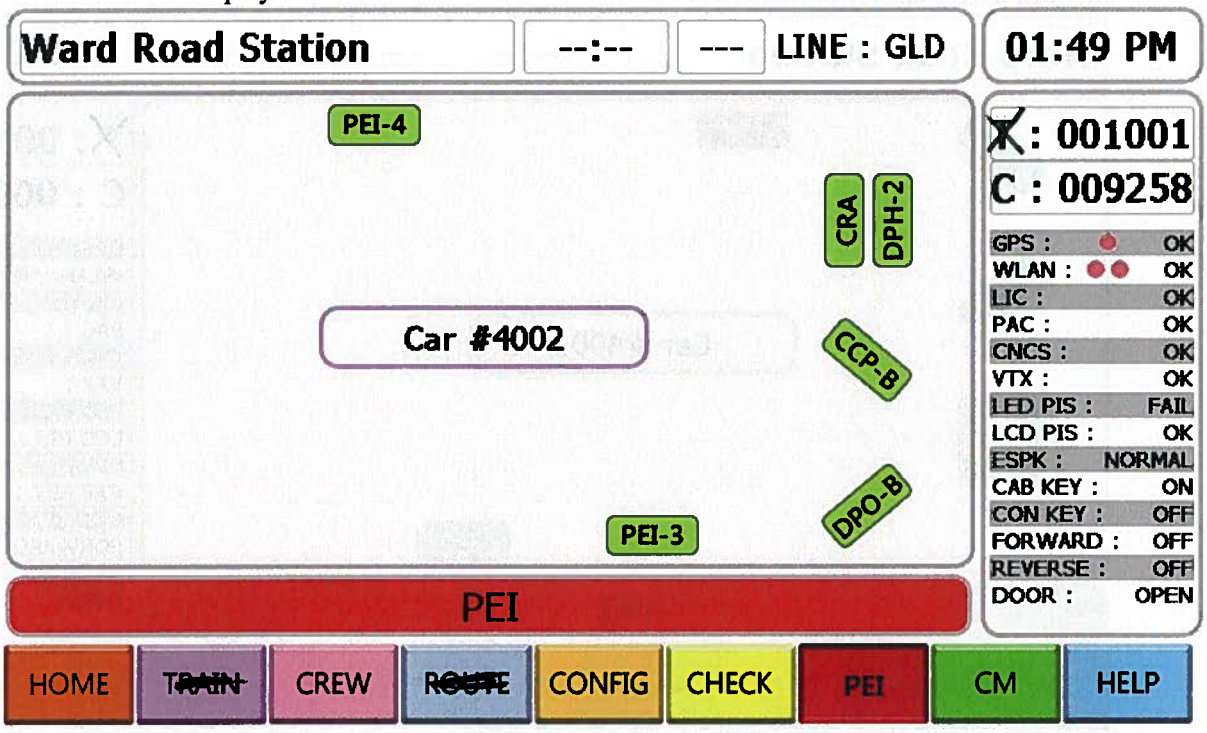
It is displayed a vehicle without PEI CALL in green and a vehicle with PEI CALL in red.

14.1 PEI CALL display of Married A

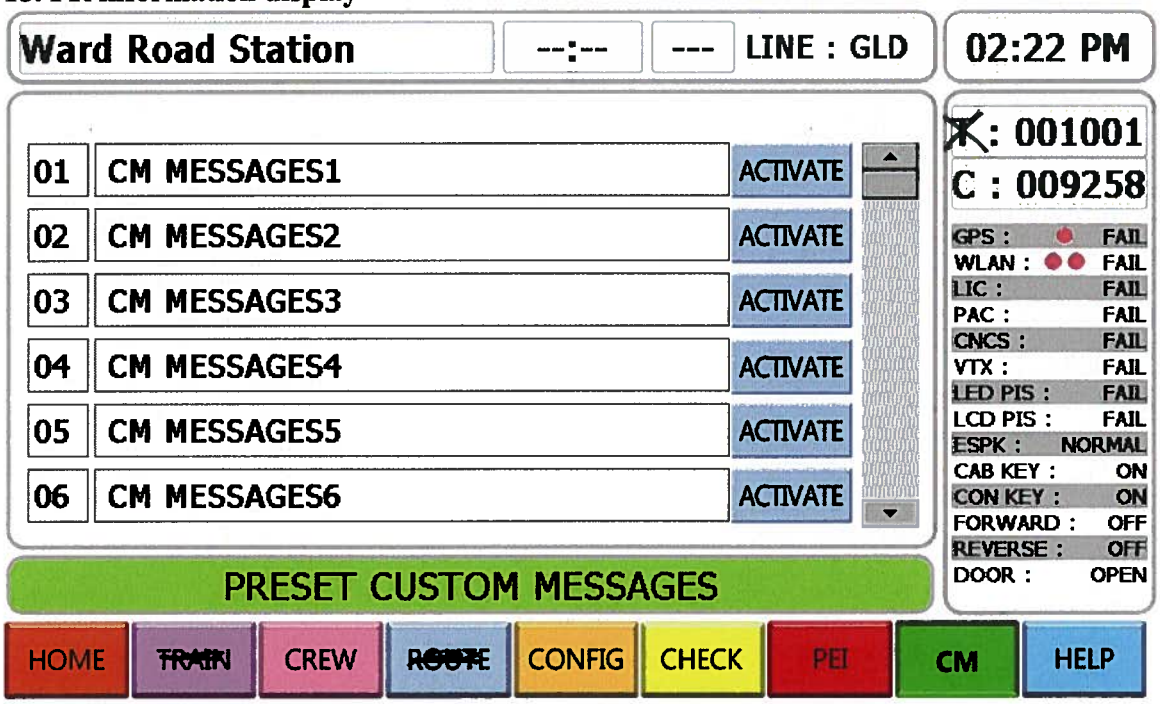


- PEI unit which has been pressed or called will be displayed in red color.

14.2 PEI CALL display of Married B

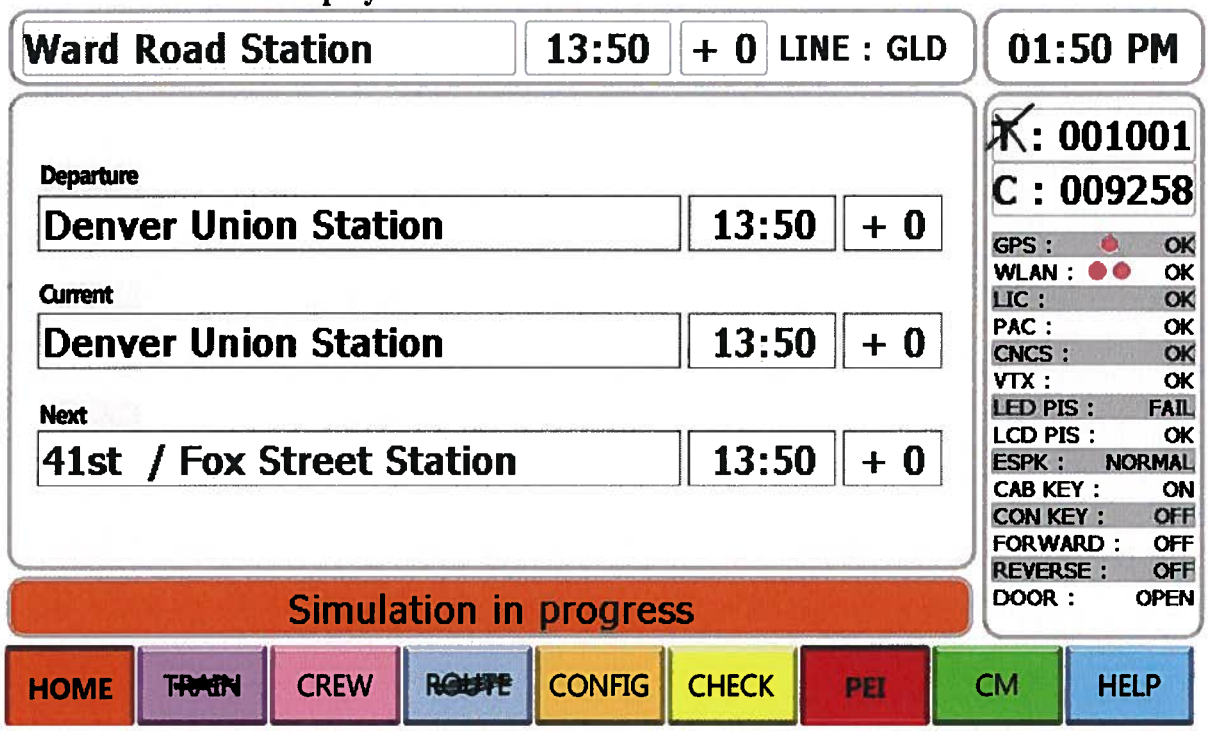


15. PR information display



- A user-selected one of listed messages will be displayed on the indicator.

16 Simulation Mode Display



#HOME

The HOME Screen is the default screen that is displayed by the CCP. ROUTE
It provides the crew with real time Train Operational & Status Data including; ~~Train~~ Number, Crew Number, ~~Route~~, Destination, ETA, Delay, Current Time, Origin, Current Station and Next Station. LINE

Additionally, this screen displays the current operational status of various Communications System devices and alerts the crew of any PEI calls.

ROUTE

#~~TRAIN~~ ROUTE

The ~~TRAIN~~ Screen allows the operator to enter a ~~TRAIN~~ ROUTE NUMBER.

The ~~TRAIN~~ ROUTE Screen can be displayed at any time by pressing the ~~TRAIN~~ ROUTE button on the CCP Menu Bar.

Upon system start-up, the HOME screen will automatically prompt the operator to enter the ~~train~~ route number.

When the ~~TRAIN~~ ROUTE Button is pressed, the CCP will display a touch panel number pad allowing for the entry of a valid six digit ~~Train~~ Route Number.

After entering the desired number, press the ENTER key to accept.

If the entered number is not valid, the CCP will indicate that the entry is invalid and will prompt the operator to enter a valid number.

If the entered ~~Train~~ Route Number is valid, the CCP will load the corresponding route schedule and initiate the Route Processing functionality.

#CREW

The CREW Screen allows the operator to enter a CREW ID.

The CREW Screen can be displayed at any time by pressing the CREW button on the CCP Menu Bar.

Upon system start-up and after the ~~Train~~ Route Number has been entered, the HOME screen will automatically prompt the operator to enter the Crew ID Number.

When the CREW button is pressed, the CCP will display a touch panel number pad allowing for the entry of a six digit CREW ID Number.

After entering the CREW ID Number, press the Enter key to accept.

The system does not check or verify the validity of the entered CREW ID Number.

The entered CREW ID is utilized by the Communications System for identification purposes within the system activity log files.

STOPS

#~~ROUTE~~ STOPS

The ~~ROUTE~~ Screen allows the operator to set several attributes pertaining to each station included within the defined route for the currently selected ~~Train~~ Route Number.

The ~~ROUTE~~ STOPS Screen can be displayed at any time by pressing the ~~ROUTE~~ STOPS button on the CCP Menu Bar.

When the ~~ROUTE~~ STOPS button is pressed, the CCP will display a detailed Route Editing screen. A scroll bar is provided allowing the operator to scroll through all of the defined stations within the route. The operator can set the following attributes for each station within the displayed station list:

1) STOP/SKIP: Each station within a defined ~~Train~~ Route Number route will default to the state established within the Denver RTD master schedule. The operator can toggle the parameter for any station between STOP and SKIP by pressing the STOP/SKIP buttons.
   The current state will be highlighted and stored into the current operating parameter table. If the parameter is set to STOP, the Communications System will initiate all AVL messages associated with the specified station for the selected route.
   If the parameter is set to SKIP, the Communications System will ignore all AVL messages associated with the specified station for the selected route.

2) EXP/ LOCAL: Each station within a defined ~~Train~~ Route Number route will default to the state established within the Denver RTD master schedule.
   The operator can toggle the parameter for any station between EXPRESS and LOCAL by pressing the EXP/LOCAL buttons.
   The current state will be highlighted and stored into the current operating parameter table.
   If the parameter is set to EXP, then the Communications System will consider this to be an EXPRESS Train Route and will ignore all AVL messages associated with the specified stations between the current station and the station designated as EXP.
   If the parameter is set to LOCAL, the Communications System will initiate all AVL messages associated with the specified station for the selected route.

#CONFIG

The CONFIG Screen allows the operator to review and set several user defined Communications System operating parameters.
The CONFIG Screen can be displayed at any time by pressing the CONFIG button on the CCP Menu Bar.

The following parameters can be set via the CONFIG Screen:

1) SPEAKER VOLUME:

a) CAB SPEAKER: There are two audio output settings for the cab speakers. By pressing the appropriate button, the Communications System can be set to NORMAL or SOFT cab speaker audio output.

i) NORMAL: Sets the cab speaker volume to the default setting.
ii) SOFT: Sets the cab speaker volume to a setting that is a predefined percentage less than the default setting.

b) INTERIOR SPEAKERS: There are two audio output settings for the interior speakers. By pressing the appropriate button, the Communications System can be set to NORMAL or SOFT interior speaker audio output.

i) NORMAL: Sets the interior speakers volume to the default setting.
ii) SOFT: Sets the interior speakers volume to a settings that is a predefined percentage less than the default setting.

c) EXTERIOR SPEAKERS: There are three audio output settings for the Exterior Speakers.
By pressing the appropriate button the Communications System can be set to NORMAL, SOFT, or MUTE.

i) NORMAL: Sets the exterior speakers volume to the default setting.
ii) SOFT: Sets the exterior speakers volume to a settings that is a predefined percentage less than the default setting.
iii) MUTE: Sets the exterior speakers volume to OFF.
This will prevent Automated Announcements from being heard on the Exterior Speakers.

2) LCD DIMMER:

a) Use the slide bar to set the CCP LCD screen dimming level.
By sliding the bar in either direction, the LCD brightness can be adjusted from 20–100% of available brightness.
b) The number at the center of the CONFIG / LCD DIMMER screen indicates the currently selected setting as a percentage of total available brightness.

3) SYSTEM RESET:

a) By pressing the SYSTEM RESET button, ALL operator adjustable Communications System settings are reset to their default settings.
b) An automatic reset of the Communications System settings also occurs after complete power down and reboot of the system, or when a new ~~Train~~ Route Number is entered.

4) SIMULATION

The SIMULATION screen allows the setting of the automatic announcement time period during Simulation mode to 10, 20 or 30 seconds.

5) C/I SETTING

The C/I SETTING screen allows the operator to identify and revise the Converter/Inverter (C/I) phase angle order of all connected vehicles in the train consist.

The C/I SETTING screen displays, for all connected vehicles (maximum of 8), the C/I phase angle order and vehicle number information.

The C/I order may be manually changed by the operator by selecting a vehicle icon, changing the order of the vehicle using the number pad and pressing the APPLY button.

#CHECK

The CHECK screen allows the operator to check the status of each car in the connected train consist, as well as visually inspect the physical devices of the Communications System in each car to confirm that each is functioning properly.

In the event that a critical component is in a failure state, the fault will be displayed on the CCP for the operator to review.

1) CAR STATUS:
   a. Press the CHECK button to navigate to the Consist Overview page, which allows the operator to see what car numbers make up the current train consist.
   b. If there is an error or fault on any Communications System component, the Icon for the car number that has the error or fault will be illuminated yellow.

2) DEVICE STATUS:
   a. Press the Car Number Icon button to navigate to the Device Status screen corresponding to that car number.
   b. The Device Status screen allows the crew to view the current status for ALL Communications System components monitored on the selected car.
   c. Component icons in red indicate an error or fault.

3) SIMULATION:
   a. The SIMULATION button allows the crew to simulate the currently selected Route Trigger Points in order to test the content playlist and verify that all information displays correctly on all Passenger Information Displays and Destination Indicators, as well as verifying that PA System Automated Announcements are working properly.

#PEI

The PEI screen allows the operator to see where in the train consist an Active PEI Call was initiated from. Car Number and PEI Box ID are displayed.

a. When a PEI Call is initiated by a passenger, the CCP in the active cab will display the following message:
"Passenger Emergency Intercom #XXXX Call" (XXXX = Car Number). This message provides the operator with an alert that a PEI Call has been initiated by a passenger along with the location of the PEI Call.
The following Automated PA Announcement in the cab will be heard each time a PEI Call is initiated by a passenger:
"A passenger needs assistance in Car Number XXXX." (XXXX = Car Number)

b. The PEI Screen, which displays where in the train consist the PEI Call was initiated, can be accessed by pressing the PEI button on the CCP menu bar.

c. The specific PEI Icon that is displayed in red indicates an active PEI Call.

#CM

The CM screen allows the operator to send custom preset messages to the LED and LCD displays.

a. Pre-defined messages will be listed on the CM screen. The operator is able to view the possible custom message options by using the scroll bar.
b. A selected message will be immediately sent to LED and LCD displays when the operator presses the ACTIVATE button next to one of the custom present messages.

Only one message activation is available at one time. Any previous message will be cancelled when activating a new custom preset message.